COSC 42: Distributed Computing

Problem 1. ***n*-Process Locks from Two-Process Locks**.
Reduce n-process locks to two process locks. That is, let $n \geq 2$ be any fixed and known integer, and let L_2 be a solution to the mutual exclusion problem for two-processes, construct by reduction a lock L_n that solves the mutual exclusion problem for n-processes. Your reduction should only use the read and write shared-memory instructions (i.e., it should not use RMW instruction) and it should satisfy the following requirements:

(LLF) If L_2 satisfies Mutex and Live-Lock Freedom but not Starvation Freedom for two processes, then L_n should satisfy Mutex and Live-Lock Freedom but not Starvation Freedom for n processes.

(SF) If L_2 satisfies Mutex and Starvation Freedom for two processes, then L_n should satisfy Mutex and Starvation Freedom for n processes.

You must design the reduction and prove that it satisfies the requirements (LLF) and (SF). You may assume that n is fixed and known and that the processes are named $1, 2, \ldots, n$.

Solution.